

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 42

Level 3 — Advanced · Number Sense II

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 3 — Advanced	Topic Focus	Number Sense II — Factors, Multiples, HCF & LCM
Questions	20	Time Allowed	40 minutes
Total Marks	20	Worksheet No.	42 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 42 introduces HCF and LCM through their real meanings — the bell/bus 'when do they meet again?' (LCM) and the 'longest equal piece' / 'most packets' problems (HCF). It also sharpens prime reasoning, factor-counting, co-prime pairs, and the digit-sum test for divisibility by 9. This is core Advanced number theory for Class 4 olympiads.

Learning Objectives

- Find HCF and LCM of small numbers by listing.
- Apply LCM to 'ring/leave together again' problems.
- Apply HCF to 'longest equal pieces' and 'most packets' problems.
- Identify primes, co-prime pairs, and count factors.
- Use the digit-sum test for divisibility by 9.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

**SECTION A · Multiple Choice Questions · 20 Questions**

- Q1.** What is the SMALLEST number that is a multiple of BOTH 4 and 6 (the LCM)?
(A) 10 (B) 12
(C) 24 (D) 8
- Q2.** What is the GREATEST number that divides BOTH 18 and 24 exactly (the HCF)?
(A) 3 (B) 6
(C) 9 (D) 12
- Q3.** Which number is PRIME?
(A) 21 (B) 27
(C) 29 (D) 33
- Q4.** How many PRIME numbers are there between 1 and 10?
(A) 3 (B) 4
(C) 5 (D) 6
- Q5.** Two bells ring together at 8:00 a.m. One rings every 6 minutes, the other every 8 minutes. When do they NEXT ring together?
(A) 8:18 a.m. (B) 8:24 a.m.
(C) 8:48 a.m. (D) 8:30 a.m.
- Q6.** The number 36 can be written as a product of two numbers. Which pair is NOT correct?
(A) 4×9 (B) 6×6
(C) 3×12 (D) 8×5
- Q7.** Which number has EXACTLY 3 factors?
(A) 6 (B) 8
(C) 9 (D) 12
- Q8.** The first common multiple of 5 and 9 is:
(A) 14 (B) 40
(C) 45 (D) 90
- Q9.** 48 sweets and 36 toffees are packed so each packet has the SAME number of sweets and the SAME number of toffees, with the GREATEST possible items per packet matched. The greatest number of packets is:
(A) 6 (B) 12
(C) 4 (D) 18
- Q10.** Which of these is a multiple of 3 AND of 5 AND of 2?
(A) 15 (B) 20
(C) 30 (D) 45

Q11. What is the sum of the first 5 multiples of 4?

- (A) 40 (B) 50
(C) 60 (D) 30

Q12. A number is a factor of 30 and also a multiple of 5. Which CANNOT be that number?

- (A) 5 (B) 10
(C) 15 (D) 20

Q13. Two ribbons of length 12 m and 18 m are cut into pieces of EQUAL length, as long as possible with no waste. The length of each piece is:

- (A) 3 m (B) 6 m
(C) 4 m (D) 9 m

Q14. Which statement is TRUE?

- (A) All even numbers are composite (B) 2 is the only even prime number
(C) 1 is a prime number (D) All odd numbers are prime

Q15. The LCM of 3, 4 and 6 is:

- (A) 6 (B) 12
(C) 24 (D) 18

Q16. A gardener plants trees in rows. He can make rows of 8 or rows of 12 with none left over. The SMALLEST number of trees he could have is:

- (A) 20 (B) 24
(C) 48 (D) 96

Q17. How many factors does the number 16 have?

- (A) 3 (B) 4
(C) 5 (D) 6

Q18. Which pair of numbers has an HCF of 1 (no common factor except 1)?

- (A) 8 and 12 (B) 9 and 16
(C) 6 and 15 (D) 10 and 20

Q19. A number leaves remainder 0 when divided by 9. Which MUST also be true?

- (A) Its digits add up to a multiple of 9 (B) It is even
(C) It ends in 9 (D) It is a prime number

Q20. Buses to two towns leave a stand: one every 15 minutes, the other every 20 minutes, starting together at 9:00 a.m. They next leave together at:

- (A) 9:35 a.m. (B) 9:60 a.m.
(C) 10:00 a.m. (D) 9:45 a.m.

ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	B	C	B	B	D	C	C	B	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	D	B	B	B	B	C	B	A	C

Step-by-Step Solutions

Q1. Correct answer: (B)

Multiples of 4: 4, 8, **12**... Multiples of 6: 6, **12**... The smallest common one is **12**.

Q2. Correct answer: (B)

Common factors of 18 and 24: 1, 2, 3, 6. The greatest is **6**.

Q3. Correct answer: (C)

29 has only factors 1 and 29. ($21 = 3 \times 7$, $27 = 3 \times 9$, $33 = 3 \times 11$.)

Q4. Correct answer: (B)

2, 3, 5, 7 → **4** primes. (1 is not prime.)

Q5. Correct answer: (B)

LCM of 6 and 8 = 24. They ring together again after 24 minutes → **8:24 a.m.**

Q6. Correct answer: (D)

$8 \times 5 = 40$, not 36. So **8×5** is wrong. (4×9 , 6×6 , 3×12 all equal 36.)

Q7. Correct answer: (C)

9 has factors 1, 3, 9 — exactly 3. (A number has exactly 3 factors only if it is the square of a prime.)

Q8. Correct answer: (C)

LCM of 5 and 9 = $5 \times 9 =$ **45** (they share no common factor).

Q9. Correct answer: (B)

The number of packets = HCF of 48 and 36 = **12** (each packet: 4 sweets and 3 toffees).

Q10. Correct answer: (C)

A multiple of 2, 3 and 5 must be a multiple of 30. Only **30** fits.

Q11. Correct answer: (C)

$4 + 8 + 12 + 16 + 20 =$ **60**.

Q12. Correct answer: (D)

Factors of 30 that are multiples of 5: 5, 10, 15, 30. **20** is not a factor of 30, so it cannot be.

Q13. Correct answer: (B)

Longest equal length = HCF of 12 and 18 = **6 m**.

Q14. Correct answer: (B)

2 is the only even prime. (2 is even but prime — making 'all even numbers are composite' false; 1 is neither prime nor composite; 9 is odd but not prime.)

Q15. Correct answer: (B)

Smallest number divisible by 3, 4 and 6: **12** ($12 \div 3 = 4$, $12 \div 4 = 3$, $12 \div 6 = 2$).



Q16. Correct answer: (B)

Smallest number divisible by both 8 and 12 = LCM = **24**.

Q17. Correct answer: (C)

Factors of 16: 1, 2, 4, 8, 16 → **5** factors.

Q18. Correct answer: (B)

9 and 16 share no factor except 1 ($9 = 3 \times 3$, $16 = 2 \times 2 \times 2 \times 2$). Such numbers are 'co-prime'.

Q19. Correct answer: (A)

A number is divisible by 9 exactly when **its digits add up to a multiple of 9** (e.g. $153 \rightarrow 1+5+3 = 9$).

Q20. Correct answer: (C)

LCM of 15 and 20 = 60 minutes. Next together = $9:00 + 60 \text{ min} = \mathbf{10:00 \text{ a.m.}}$

